Track B — Avionics and Communications

RescueSwarm: work package and funding request

NIDAR 2026–27 · Track 1 · Mission 1

Ask: INR 280,963 · 1–2 students · Flight controller, companion computer, GNSS/RTK, mesh, safety link, video

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This track owns everything electrical between the airframe and the autonomy. It carries the second-largest ask in the programme, and it carries the single component whose specification is currently in doubt.

1 Scope and ownership

This track owns **flight controller, companion computer, gnss/rtk, mesh, safety link, video**. It is one of four delivery tracks defined in `docs/development-plan.md` §1.3; the integrated system argument lives in the master proposal, `docs/proposal/rescueswarm-proposal.pdf`, which this document does not restate.

2 Compute, and why it is three parts rather than one

The stack is a flight controller, a companion computer and an accelerator, and the division is not arbitrary. The accelerator is an M.2 class device whose only interface is **PCIe**; it is a neural processing unit, not a computer, and it cannot boot, hold a filesystem or terminate a camera link. The camera is a CSI module producing a Bayer-mosaic stream that is not an image until something demosaics and scales it. The companion computer is what both plug into.

This has a hard consequence for procurement. The accelerator requires a host that exposes PCIe. That rules out the previous-generation single-board computer entirely — its PCIe lane is committed internally to a USB controller and is never broken out. There is no adapter and no workaround. The host generation is a dependency of the accelerator, not a performance preference, and any cost pass that treats it as one produces a bill of materials that cannot be assembled.

3 Positioning: the item this track must resolve first

Geolocation accuracy governs **125 of 200** available geotagging points, and the error budget is unambiguous about what drives it:

Configuration	RSS error
Standard GNSS, uncalibrated boresight	4.57 m
Standard GNSS, calibrated boresight	3.88 m
RTK + dual-antenna heading + calibration	0.75 m
+ 20-frame fusion	0.66 m

The step from 3.88 m to 0.75 m is RTK. Nothing else in the budget moves the number comparably, which is why the receiver and its base station survive every cost-reduction pass.

The open item. The receiver currently carried against this line is specified by its supplier as supporting SBAS correction services with a CEP below 1.5 m. That is assisted GNSS, not RTK, and it lands the system in the 3.88 m row rather than the 0.75 m row. Resolving this — by confirming the part, substituting a genuine RTK receiver, or re-scoring the geotagging expectation — is this track's first P1 action, because 125 points depend on which row is true.

4 The radio link, and a deferral that depends on a configuration

The sub-GHz safety radio is deferred on the strength of a specific claim: that the primary radio link runs in a firmware mode carrying both control and MAVLink telemetry on one link and one autopilot serial port. The older transparent-serial mode *consumes* that link — configured that way the aircraft has telemetry and no control, and the deferred radio is required again.

This is not a preference, it is the precondition of a deferral, and it must be verified on the bench in P2 rather than assumed.

The mesh link budget is thin and this track should say so plainly: 5.8 GHz carries **1.7 dB** of margin at the 600 m design range against a 15 dB target for a mobile airborne link. The mitigation is architectural rather than financial — keep the ground antenna elevated, stream one switched video feed rather than three, and accept that video degrades before command does. Total offered load is 2.5 Mbps against a 20 Mbps channel, so the constraint is margin, not bandwidth.

5 Failsafe configuration

The battery failsafe thresholds are derived quantities, not chosen ones: `BATT_LOW_VOLT` 18.48 V and `BATT_CRT_VOLT` 17.18 V follow from the cell's resting curve and a pack resistance of `BATT_RESISTANCE` = 0.0400 Ω. A previous configuration set the sag-compensated voltage source while leaving pack resistance unset, so the compensation did nothing and a loaded voltage was compared against a threshold taken from the resting curve. That class of defect — a correct-looking configuration that nothing actually applies — is what this track's bench verification exists to catch.

6 Funding request

Every figure below is generated from `docs/proposal/figures/track_budget.py`, which partitions the master competition budget. The four track asks plus the shared pool reconcile to the master figure exactly; that reconciliation is asserted in code and fails the build if it ever stops holding. **K** marks a line held at professional grade on purpose.

Item		**INR**
Flight controller (1 x 22,600 x 3 ac)	**K**	67,800
GNSS RTK primary (1 x 18,000 x 3 ac)	**K**	54,000
RC rx, storage, cooling, mounts (1 x 8,500 x 3 ac)		25,500
Companion computer (1 x 8,000 x 3 ac)	**K**	24,000
Mesh node + antennas (1 x 6,000 x 3 ac)		18,000
RTK base receiver		18,000
Safety-pilot transmitter		8,000
Survey tripod + tribrach		4,000
Subtotal		219,300
Customs duty and freight		10,217
GST		14,799
Contingency at 15%		36,647
TRACK B ASK		**280,963**

Deferred or institutionally held, and therefore not requested: Spare GNSS, Spare compute module, Spare flight controller, VEGA co-processor (1 x 0 x 3 ac), WPC / ETA licensing. These remain capabilities the track requires; they are simply not being bought. Where a deferral carries a technical consequence it is stated in the risk table below rather than left implicit.

7 Deliverables by phase

Phase	Dates	This track delivers
P1	24 Aug – 13 Sep	GNSS specification resolved. Long-lead orders: compute, camera, radios, GNSS. Interface control documents agreed.
P2	14 Sep – 4 Oct	Radio mode verified on the bench. Failsafe thresholds validated against a real pack discharge.
P3	5 Oct – 18 Oct	Link range verified in the field. RTK fix demonstrated.
P5	9 Nov – 22 Nov	All links exercised under mission load.

8 Risks owned by this track

Risk	Sev.	Mitigation
Primary GNSS is not RTK-capable	High	Resolve at P1. If confirmed, either substitute a true RTK receiver or restate the geotagging expectation. 125 points ride on this.
Radio mode does not carry control and telemetry together	High	Verify on the bench at P2. If it does not, the deferred sub-GHz safety radio must be restored, which is a budget event.
5.8 GHz margin is 1.7 dB against a 15 dB target	Medium	Elevated ground antenna, single switched video feed. Command link is on a separate budget with 12.2 dB.

9 Interfaces to other tracks

With	Dependency
Track A	Depends on regulated power and on vibration-isolated mounting.
Track C	Supplies the companion computer the autonomy runs on, and the MAVLink routing it commands through.
Track D	Supplies the camera interface, the accelerator, and the time and attitude reference every geotag is computed against.

Interfaces are where student programmes fail, because each side assumes the other owns the gap. Every row above is a two-way commitment and should be recorded as an interface control document at P1.

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